Find missing dimensions

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ___________________

1. Read the model: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height. Copy or complete the first step.

2. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

3. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height. Copy or complete the first step.
   Answer 6 cm.

2. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
   Answer Both are 240 cm3.

3. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer 240 cubes.

4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.
   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

Find missing dimensions

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

2. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

3. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

4. Solve this independently and show every stage: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

5. Use a second method or representation for: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

6. Check this result using an inverse, estimate or known fact: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

7. Prove or explain your reasoning: A solid has 5 layers of 24 unit cubes. What is its volume?

8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

Teacher answers and checking prompts

1. A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer 6 cm.

2. Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
   Answer Both are 240 cm3.

3. A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer 240 cubes.

4. Solve this independently and show every stage: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer 6 cm.

5. Use a second method or representation for: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
   Answer Expected result: Both are 240 cm3. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer Expected result: 240 cubes. A valid check must be shown.

7. Prove or explain your reasoning: A solid has 5 layers of 24 unit cubes. What is its volume?
   Answer 120 cubic units. The explanation must show why the method works.

8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.
   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Find missing dimensions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: ___________________

1. Prove or explain your reasoning: A solid has 5 layers of 24 unit cubes. What is its volume?

2. Prove the answer to this question, rather than only calculating it: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.

3. Solve this in two different ways and compare them: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.

4. Work backwards from the answer to reconstruct the question: 240 cubes.

5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.

6. Create a new example that tests the same idea as: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A solid has 5 layers of 24 unit cubes. What is its volume?
   Answer 120 cubic units. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A cuboid has volume 360 cm3, length 12 cm and width 5 cm. Find its height.
   Answer Expected result: 6 cm. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
   Answer Expected result: Both are 240 cm3. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 240 cubes.
   Answer One valid reconstruction is: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.
   Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

6. Create a new example that tests the same idea as: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.